Neural Network Weight Update Calculation

Calculation Report

November 2025

1 Network Architecture

1.1 Initial Parameters

- **Target (y_{target}):** 0.5.
- **Learning Rate (η):** 0.1.
- **Activation Function:** $f(x) = \frac{1}{1+e^{-x}}$.
- **Weights to Update:** $W_{13} = 0.3, W_{23} = 0.9$.

2 Calculation Process

2.1 Iteration 1

1. Forward Propagation
Hidden layer outputs (pre-calculated): $y_1 \approx 0.68, y_2 \approx 0.66$.
Output summation:

$$S_{out} = (y_1 \cdot W_{13}) + (y_2 \cdot W_{23}) = (0.68 \cdot 0.3) + (0.66 \cdot 0.9) = 0.204 + 0.594 = 0.801$$

Prediction (y_{pred}):

$$y_{pred} = \frac{1}{1 + e^{-0.801}} \approx \mathbf{0.69}$$

2. Error & Gradient

$$E = y_{target} - y_{pred} = 0.5 - 0.69 = -\mathbf{0.19}$$

Gradient (δ_3) for output layer:

$$\delta_3 = y_{pred}(1 - y_{pred}) \cdot E = 0.69 \cdot 0.31 \cdot (-0.19) \approx -\mathbf{0.0406}$$

3. Weight Update

$$\Delta W = \eta \cdot y_{in} \cdot \delta$$

$$W_{13}^{new} = 0.3 + (0.1 \cdot 0.68 \cdot -0.0406) = 0.3 - 0.0027 = \mathbf{0.2973}$$

$$W_{23}^{new} = 0.9 + (0.1 \cdot 0.66 \cdot -0.0406) = 0.9 - 0.0027 = \mathbf{0.8973}$$

2.2 Iteration 2

Using updated weights $W_{13} = 0.2973, W_{23} = 0.8973$.

1. Forward: Sum $= 0.68(0.2973) + 0.66(0.8973) \approx 0.794$.
$y_{pred} = \sigma(0.794) \approx \mathbf{0.688}$.

2. Error: $E = 0.5 - 0.688 = -0.188$.

3. Gradient: $\delta_3 = 0.688(1 - 0.688)(-0.188) \approx -0.0403$.

4. Update:

$$W_{13}^{new} = 0.2973 - 0.0027 = \mathbf{0.2946}$$

$$W_{23}^{new} = 0.8973 - 0.0027 = \mathbf{0.8946}$$

2.3 Iteration 3

Using updated weights $W_{13} = 0.2946, W_{23} = 0.8946$.

1. Forward: $y_{pred} \approx \mathbf{0.687}$.

2. Error: $E = -0.187$.

3. Gradient: $\delta_3 \approx -0.0402$.

4. Update:

$$W_{13}^{new} = 0.2946 - 0.0027 = \mathbf{0.2919}$$

$$W_{23}^{new} = 0.8946 - 0.0026 = \mathbf{0.8920}$$

2.4 Iteration 4

Using updated weights $W_{13} = 0.2919, W_{23} = 0.8920$.

1. Forward:
Sum $= 0.68(0.2919) + 0.66(0.8920) = 0.1985 + 0.5887 \approx 0.787$.
$y_{pred} = \sigma(0.787) \approx \mathbf{0.6872}$.

2. Error: $E = 0.5 - 0.6872 = -0.1872$.

3. Gradient: $\delta_3 = 0.6872(1 - 0.6872)(-0.1872) \approx -0.0402$.

4. Update:

$$W_{13}^{new} = 0.2919 + (0.1 \cdot 0.68 \cdot -0.0402) = 0.2919 - 0.0027 = \mathbf{0.2892}$$

$$W_{23}^{new} = 0.8920 + (0.1 \cdot 0.66 \cdot -0.0402) = 0.8920 - 0.0027 = \mathbf{0.8893}$$

3 Key Results (Training Log)

i (Iter)	**j (Weight)**	W_{old}	W_{new}	δ **(Gradient)**	η
1	W_{13}	0.3000	0.2973	-0.0406	0.1
1	W_{23}	0.9000	0.8973	-0.0406	0.1
2	W_{13}	0.2973	0.2946	-0.0403	0.1
2	W_{23}	0.8973	0.8946	-0.0403	0.1
3	W_{13}	0.2946	0.2919	-0.0402	0.1
3	W_{23}	0.8946	0.8920	-0.0402	0.1
4	W_{13}	0.2919	0.2892	-0.0402	0.1
4	W_{23}	0.8920	0.8893	-0.0402	0.1

Table 1: Evolution of Output Layer weights over 4 iterations.

4 Highlights

- **Error Reduction:** The error continues to decrease monotonically, reaching $|0.1872|$ in the fourth iteration.
- **Convergence:** The gradient (δ_3) has stabilized around -0.0402, indicating a consistent but slow descent down the error surface.
- **Weight Adjustment:** Both weights continue to decrease by approximately 0.0027 per step.